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Supercomputing
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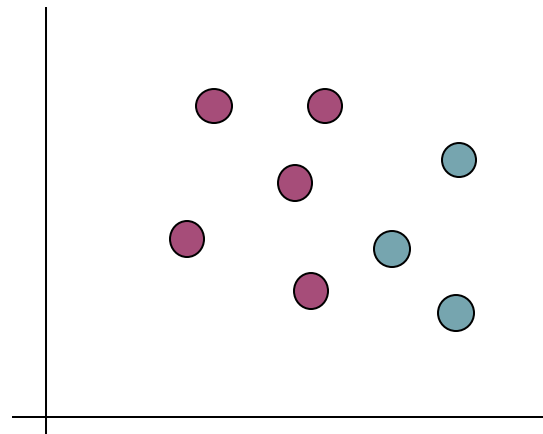


Supervised learning

A natural approach

SVMs

- ❖ A new way of classifying
 - Boundary
 - Only a few samples needed
- ❖ Worst name ever (Vapnik, 1963)



Data boundaries, Hyperplanes & Margins

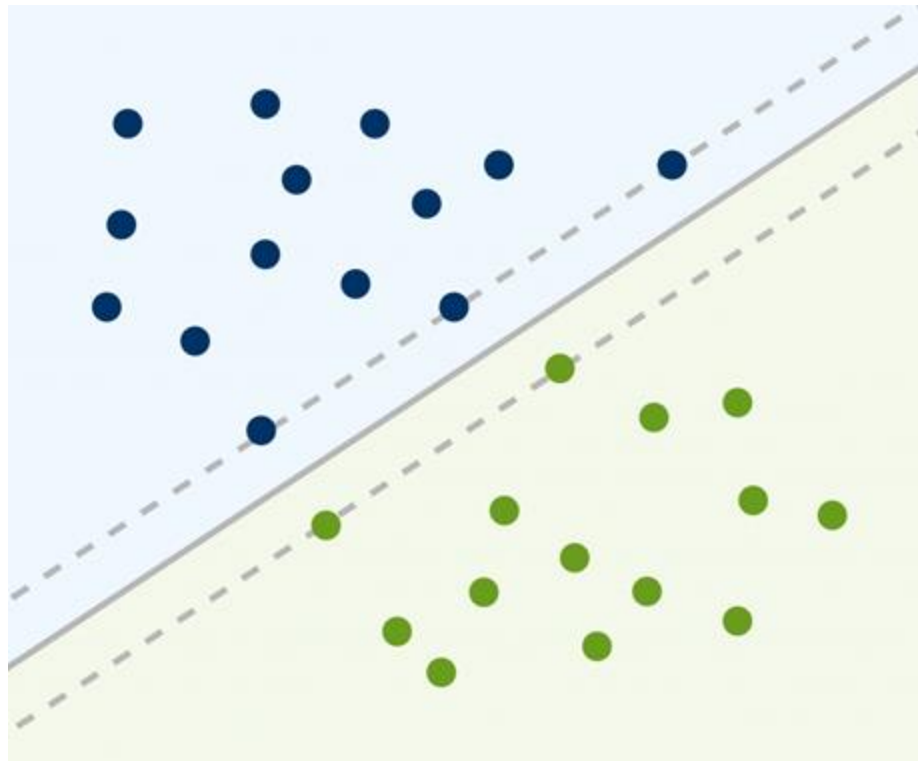
❖ Margins

- Narrower, less data, less signal
- Wider, more data, more noise

❖ A balanced optimal

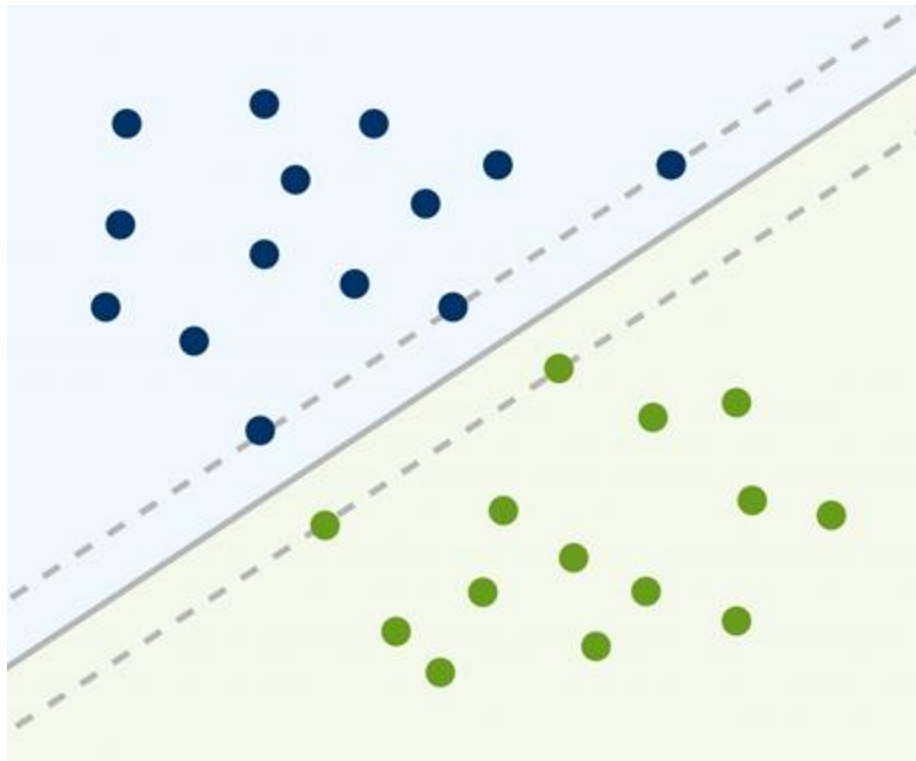
- Bias / Underfit
- Variance / Overfit

❖ Consistency



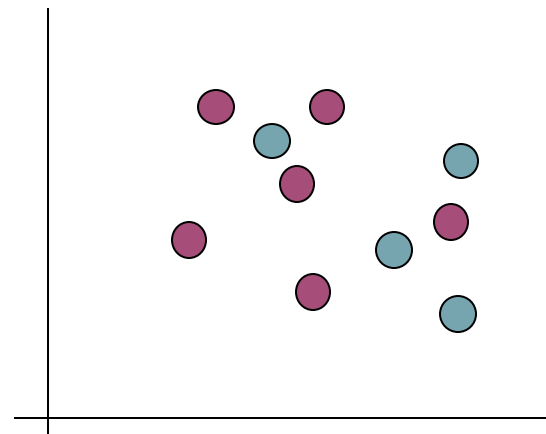
Boundaries, Hyperplanes & Margins

- ❖ Support vectors
 - Edge cases
- ❖ Optimal hyperplane
 - Max. margin
- ❖ Geometry based
 - No Gaussian assumption
- ❖ On inference, test sign



Imperfect machines

- ❖ The boundary: $w * x + b = 0$
 - What when its not perfectly separable?
- ❖ SVM has two goals
 - Find *maximal margin hyperplane*
 - Find a hyperplane which minimizes error



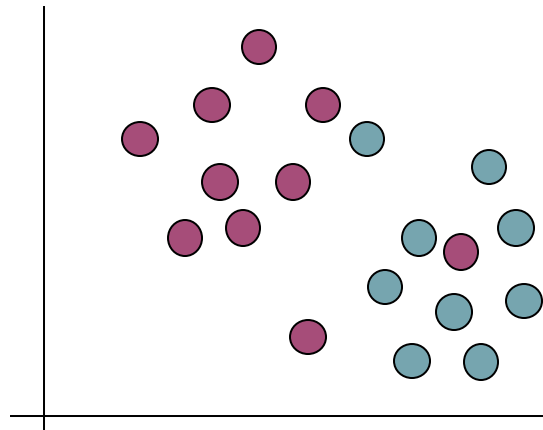
Optimization & Hyperparameters

- ❖ C (regularization param.) as a trade-off
 - Low C , margin matters more (underfit!)
 - High C , sample loss matters more (overfit!)

$$\underbrace{\frac{1}{2}\|w\|^2}_{\text{Magnitude of weight vector (margin size)}} + C \sum_i \max(0, 1 - \underbrace{y_i}_{\text{label } [-1,1]} \underbrace{(w^T x_i + b)}_{\text{data}})$$

Magnitude of weight vector (margin size)
Normal vector to the hyperplane

Error of each training sample



Cool, not cool enough

- ❖ *SVM classifier, regressor (even clustering)*
 - SVR: Replace sample error by distance (within a margin ϵ)
 - $\Sigma(\max(0, |y_i - (w * x_i + b)| - \epsilon))$
- ❖ Deterministic
- ❖ Outlier resistant
- ❖ No gaussian assumption
- ❖ Still a linear model

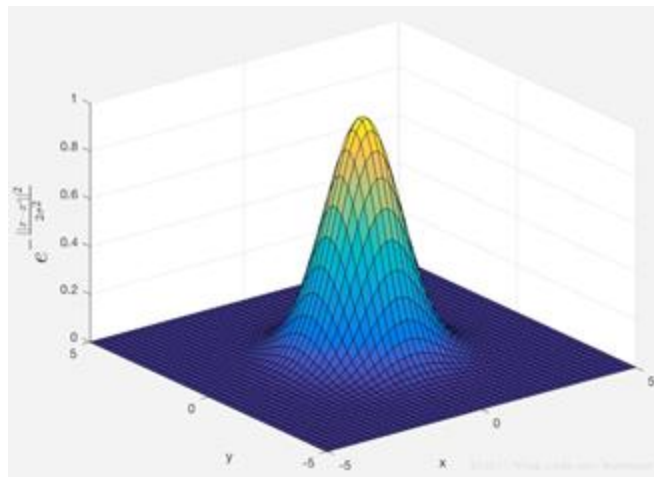
SVM beyond linearity

❖ *Kernel trick!

- ~~° Transform data~~ to non-linear, high dim. space
- Use a kernel to estimate distance in high dim. space
- Find linear hyperplane
- Transform hyperplane into low dim.

*Applicable to PCA, KNN, ...

° That's the trick!

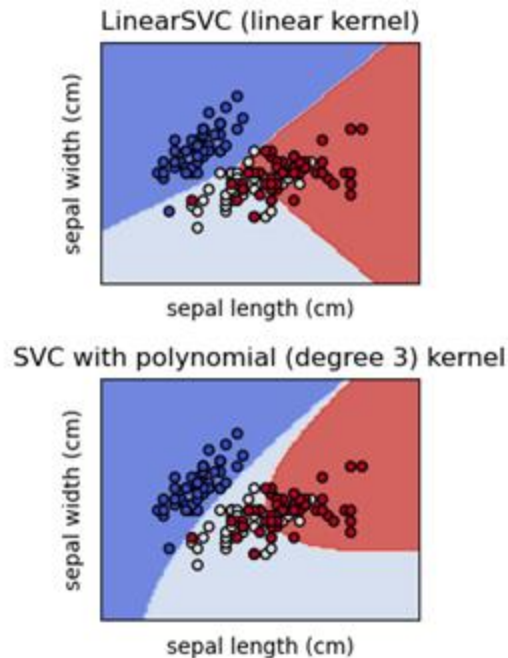
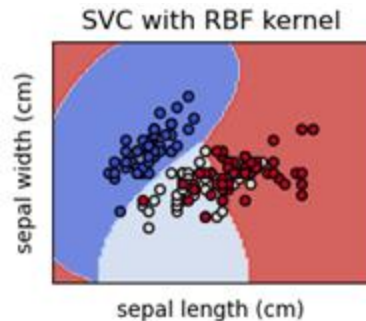


SVM beyond linearity

- ❖ Most popular: Radial Basis Function (RBF)
 - $K(x, y) = \exp(-\gamma * ||x - y||^2)$
 - Depends on radial distance, γ shape of the kernel:
 - Smaller values: Broad kernel, slow decay, smoother curve
 - Larger values: Narrow kernel, fast decay, peak-like curve
 - $||x - y||^2$ squared Euclidean distance between data points x & y .

SVM with kernels

- ❖ Much better for high-dimensional domains
 - e.g., num. features > num. samples
- ❖ Dangerous in very high-dimensional domains
 - Likely to overfit
 - Expensive

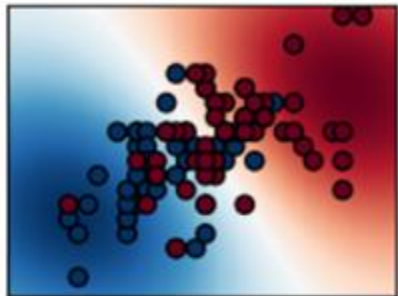


RBF radius

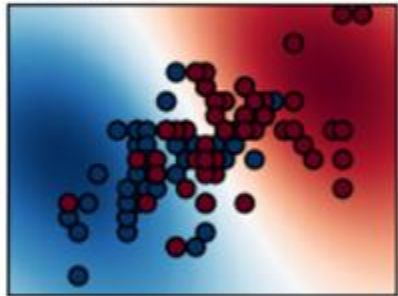
❖ Gamma

- Closer to zero, more SVs (underfit)
- Higher, less SVs (overfit)

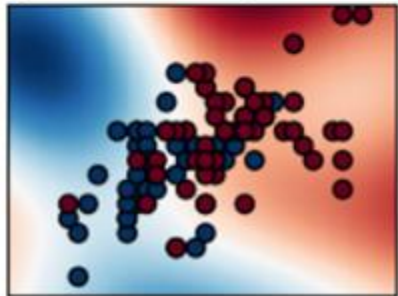
$\gamma=10^{-1}, C=10^{-2}$



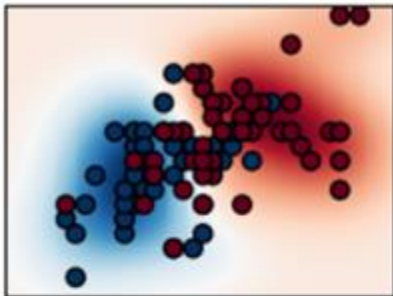
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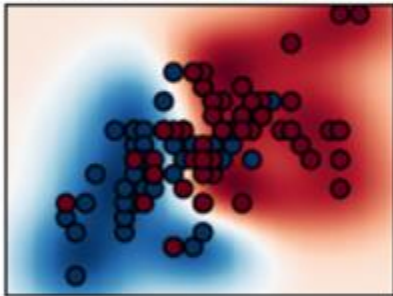
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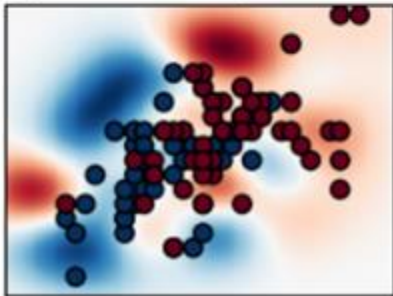
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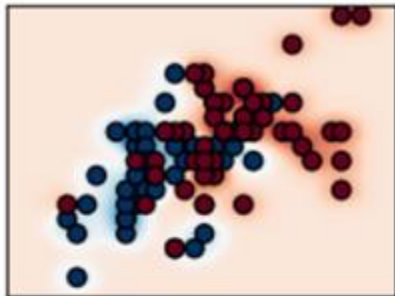
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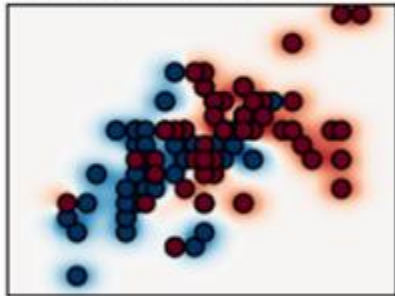
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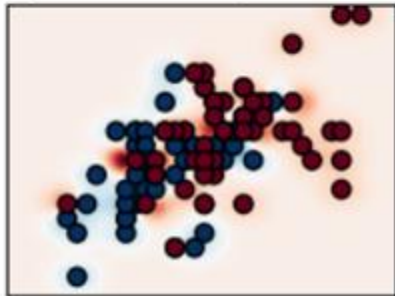
$\gamma=10^1, C=10^{-2}$



$\gamma=10^1, C=10^0$



$\gamma=10^1, C=10^2$



A thin line

default $C = 1.0$

default gamma = scaled to data variance & size

